

Introduction

The damages in Maine from the significant flooding events in December 2023 altered the livelihoods of families all over the state. As a Maine resident and an Earth and Climate Scientist, I find this issue disturbing yet intriguing. This data analysis provides a broader context for the December 2023 flooding event. It will investigate how extreme that event was while exploring whether such events are occurring more frequently. This topic is an ideal intersection of my Digital and Computational Studies Minor and my Earth and Climate Science Major. This will develop my skills in analysis and visualization in R and my understanding of river systems. Furthermore, it is a subject relevant to the lives around me, inspiring me to pursue an in-depth investigation into the topic at hand.

The Dataset

The United States Geological Survey (USGS) has implemented river gauging stations around the US for over a century. The Lewiston—Auburn Androscoggin gauge station began collecting data in January 1966. Data parameters include biological, nutrient, inorganic, and sediment. In September 1995, the station began collecting discharge data. Data is collected every 15 minutes using a flow rate gauge and river bed area calculations. These two variables allow an accurate calculation of the discharge. This consistent and accurate data provides an opportunity to investigate flooding events like this one thoroughly. The discharge and a time series will be the primary variables used. Discharge and other aforementioned parameters are all available in an easily accessible dataset provided by the USGS, 'dataRetrieval'. It is well-organized and easy to navigate. The desired gauge station and preferred parameters can easily be accessed.

Methods

```
siteNo <- "01059000"
pCode <- "00060"
start.date <- "1995-01-01"
end.date <- "2024-03-04"

AndroData <- readNWISuv(siteNumbers = siteNo,
                        parameterCd = pCode,
                        startDate = start.date,
                        endDate = end.date)

AndroData <- renameNWISColumns(AndroData)
```

After installing the appropriate packages, I accessed the desired location and parameters via the code above. The siteNo object refers to the code in the “dateRetreival” package corresponding to the Lewiston - Auburn Androscoggin gauge station. The pCode object corresponds to the code of the desired parameter; “00060” refers to discharge. Start.date and end.date are self explanatory. The code following the assignment statements is the appropriate method to create a data set with informative labels as instructed on the USGS website.

```
start_time <- "1995-10-01 04:15:00"
end_time <- "2019-12-31 18:45:00"

twenty_five_year <- subset(AndroData, dateTime >= start_time & dateTime <= end_time)

twenty_five_year_average <- twenty_five_year %>%
  mutate(Month = month(dateTime), Day = day(dateTime), Time = format(dateTime, "%H:%M:%S")) %>%
  group_by(Month, Day, Time) %>%
  summarise(average_flow = mean(Flow_Inst, na.rm = TRUE)) %>%
  mutate(DateTimeID = as.POSIXct(paste("2000", Month, Day, Time, sep = "-")))

View(twenty_five_year_average)
```

This step was done to create a baseline 25-year average to use as a standard for the discharge throughout the year. It was rather difficult to create an average discharge value for each day from the entire period due to the different years in the POSIXct format. I

executed this by isolating the time, day, and month into distinct columns from the POSIXct format. I then mutated the data set and added a column that combined the time, day, and month columns into one dateTimeID column with proper POSIXct format. I had to add an arbitrary common year for the average discharges of each date because POSIXct requires a year value. I chose "2000". Maintaining POSIXct format is critical to permit plotting date data in ggplot.

```
#Annual Flow Data 2020-2023
twenty_twenty <- AndroData %>%
  filter(dateTime >= "2019-12-31 19:00:00" & dateTime <= "2020-12-31 18:45:00") %>%
  mutate(dateTime = as.POSIXct(dateTime)) %>%
  mutate(dateTime = update(dateTime, year = 2000))

twenty_twenty_one <- AndroData %>%
  filter(dateTime >= "2020-12-31 19:00:00", dateTime <= "2021-12-31 18:45:00") %>%
  mutate(dateTime = as.POSIXct(dateTime)) %>%
  mutate(dateTime = update(dateTime, year = 2000))

twenty_twenty_two <- AndroData %>%
  filter(dateTime >= "2021-12-31 19:00:00", dateTime <= "2022-12-31 18:45:00") %>%
  mutate(dateTime = as.POSIXct(dateTime)) %>%
  mutate(dateTime = update(dateTime, year = 2000))

twenty_twenty_three <- AndroData %>%
  filter(dateTime >= "2022-12-31 19:00:00", dateTime <= "2023-12-31 18:45:00") %>%
  mutate(dateTime = as.POSIXct(dateTime)) %>%
  mutate(dateTime = update(dateTime, year = 2000))

#Plot Overlaying Average Annual Flow and Flows from 2020-2023
overlay_plot <- ggplot() +
  geom_line(data = twenty_five_year_average, aes(x = DateTimeID, y = average_flow, group = 1, color = "blue"), linewidth = 2) +
  geom_line(data = twenty_twenty, aes(x = dateTime, y = Flow_Inst, color = "green")) +
  geom_line(data = twenty_twenty_one, aes(x = dateTime, y = Flow_Inst, color = "red")) +
  geom_line(data = twenty_twenty_two, aes(x = dateTime, y = Flow_Inst, color = "yellow")) +
  geom_line(data = twenty_twenty_three, aes(x = dateTime, y = Flow_Inst, color = "aquamarine")) +
  scale_colour_manual(name = 'Year',
    values = c( 'blue' = 'blue',
                 'aquamarine' = 'aquamarine',
                 'red' = 'red', 'green' = 'green',
                 'yellow' = 'yellow'),
    labels = c( '2023', '1995-2019', '2021', '2022', '2020' )) +
  scale_x_datetime(date_breaks = "4 months", date_labels = "%b") +
  labs(title = "Instantaneous Flow over Time (Mean Values from 1995-2019)",
    x = "Date (1995 - 2019)",
    y = "Mean Instantaneous Flow",
    caption = "Data come from the USGS",
    color = "Series") # Note: 'color' label should match what the legend should represent
```

New data frames were made, isolating each of the last four full calendar years. It was quite simple. The dates were filtered and ensured they were in proper POSIXct format, and the year was changed to the arbitrary typical year chosen in the twenty-five-year average (2000). This was done so all four different calendar years and the twenty-five-year average could all be overlaid in the same POSIXct format. Each of these data frames were plotted with dates on the x-axis and the discharge on the y-axis.

$$RI = \frac{(n + 1)}{m}$$

$m = \text{rank}$
 $n = \text{total \# of years for which discharge records exist}$

```

93 #Reoccurrence_Interval w/ 2023
94
95 Reoccurrence_Interval <- AndroData %>%
96   filter(dateTime >= "1995-12-31 19:00:00", dateTime <= "2023-12-31 18:45:00") %>%
97   mutate(Year = year(dateTime)) %>%
98   group_by(Year) %>%
99   summarise(Max_Flow = max(Flow_Inst)) %>%
100  arrange(desc(Max_Flow)) %>%
101  mutate(Flow_Rank = rank(-Max_Flow)) %>%
102  mutate(RI = (28 + 1)/Flow_Rank)
103 |
104 View(Reoccurrence_Interval)

```

A new data frame was created with the reoccurrence interval involved. The reoccurrence interval is the period between a flooding event of a particular discharge. To solve for the reoccurrence interval of a certain discharge, you rank the highest data point of discharge from each year highest to lowest (line 100). Each peak discharge is ranked in that order. The reoccurrence interval is solved by the equation above. The equation was sourced from a different class.

A logarithmic model comparing the reoccurrence interval and the discharge rate was executed, and the high R² value (0.84) demonstrated a close logarithmic relationship

```

111 model_coefficients <- coef(reoccurrence_model)
112 intercept <- model_coefficients[1] # Intercept
113 slope <- model_coefficients[2]     # Slope of the logarithm of x
114
115 summary(model_coefficients)
116 equation <- sprintf("Flow_Inst = %.2f + %.2f*log(RI)", intercept, slope)
117 print(equation)
118 # "Flow_Inst = 35627.12 + 9328.25*log(RI)"
119 equation_final <-
120 ~   function(RI) {
121 ~     35627.12 + 9328.25*log(RI)}

```

between the two variables. It is typical to have a logarithmic relationship between the reoccurrence interval and flow rate in all rivers, which is the rationale behind choosing a logarithmic relationship. Furthermore, this is why the x-axis was plotted on a logarithmic scale (line 142). Using this model, the equation for the line of best fit was solved independently by solving for the y-intercept and the slope, and the equation was merged in the function(RI). This equation is effective for solving the reoccurrence interval for any given discharge. This was plotted with the logarithmic line of best fit displayed as well.

```

148 reoccurrence_plot <- ggplot(Reoccurrence_Interval, aes(RI, Max_Flow)) +
149   geom_point() + stat_function(fun = equation_final, geom = "line", color = "blue") +
150   scale_x_continuous(trans = "log10", labels = scales::number_format()) + xlim(0,100) +
151   labs(title = "Reoccurrence Interval of Each Discharge Level",
152        x = "Reoccurrence Interval", y = "Discharge (ft^3/s)",
153        caption = "Data come from the USGS")

```

```

153 #Duration Curve
154 total_values <- 968014
155
156 Duration_Curve_data <- AndroData %>%
157   mutate(Flow_Inst_Rank = rank(desc(Flow_Inst))) %>%
158   arrange(Flow_Inst_Rank) %>%
159   mutate(duration_value= Flow_Inst_Rank/(1+ total_values))
160
161 duration_curve_plot <- ggplot(Duration_Curve_data, aes(duration_value, Flow_Inst)) +
162   geom_line() +
163   labs(title = "Duration Curve",
164         x = "Percent of Time Indicated Discharge Was Met or Exceeded",
165         y = "Discharge (ft^3/s)", caption = "Data come from the USGS") +
166   scale_x_continuous(trans = "log10", labels = scales::number_format())

```

$$P = 100 \frac{m}{n + 1}$$

P = % of time a given flow is equaled or
 exceeded (daily or annual flow data used).
 m = rank of discharge (1 = highest flow)
 n = total number of data values.

A duration curve displays the relative frequency of each discharge. The equation above solves this problem. Line 157 creates a new column to the new data frame corresponding to the rank of each discharge datapoint. Then, a new column was created with the P value solved in line 159. The total number of data points is 968014. The x-axis of the plot is on a logarithmic scale to better demonstrate the rarity of extreme flooding events. It shows very low percentages, like 0.00001.

```

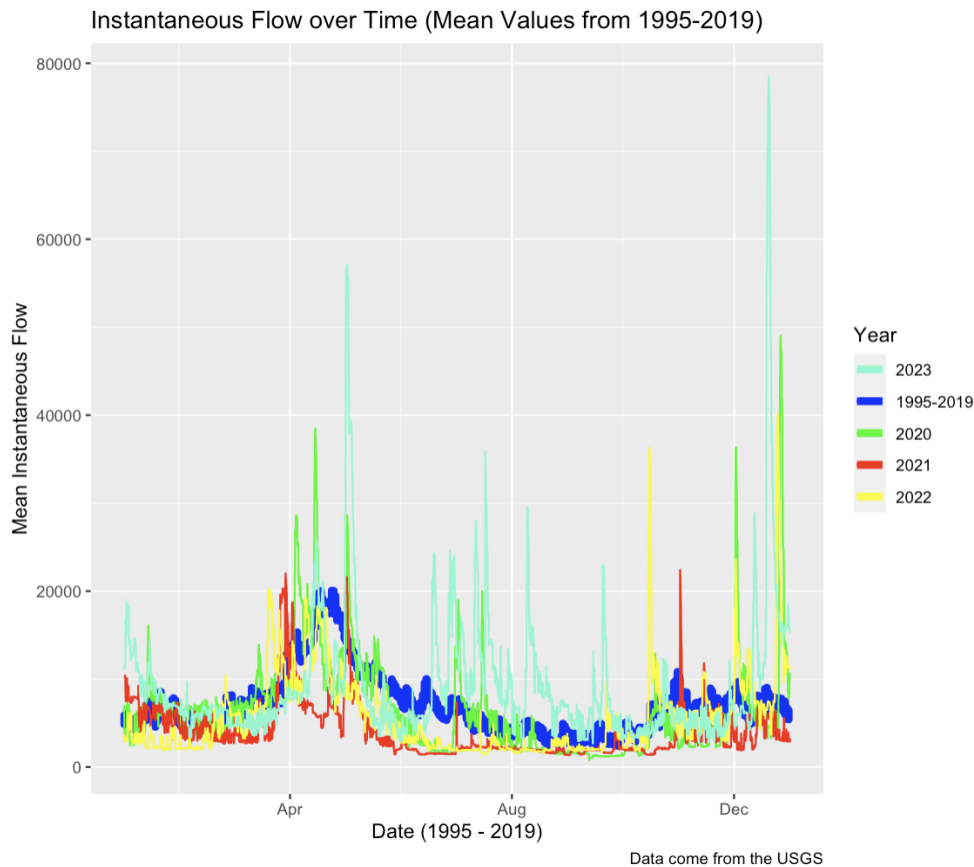
170 #Flood Threshold Surpassing
171 upper_threshold <- quantile(twenty_five_year$Flow_Inst, 0.995)
172 print(ninety_fifth_percentile)
173 #99.5th percentile = 39200
174
175 flooding_events_df <- AndroData %>%
176   filter(Flow_Inst > upper_threshold)
177
178 flooding_events_plot <- ggplot(flooding_events_df, aes(dateTime, Flow_Inst)) +
179   geom_point() +
180   labs(title = "Flooding Events Exceeding 39200 cfs 1995 - Present ",
181         x = "Date", y = "Discharge (ft^2/s)",
182         caption = "Data come from the USGS")
183
184 print(flooding_events_plot)

```

```
flooding_events_plot <- ggplot(flooding_events_df, aes(dateTime, Flow_Inst)) +  
  geom_point() +  
  labs(title = "Flooding Events Exceeding 39200 cfs 1995 - Present ",  
        x = "Date", y = "Discharge (ft^2/s)",  
        caption = "Data come from the USGS")
```

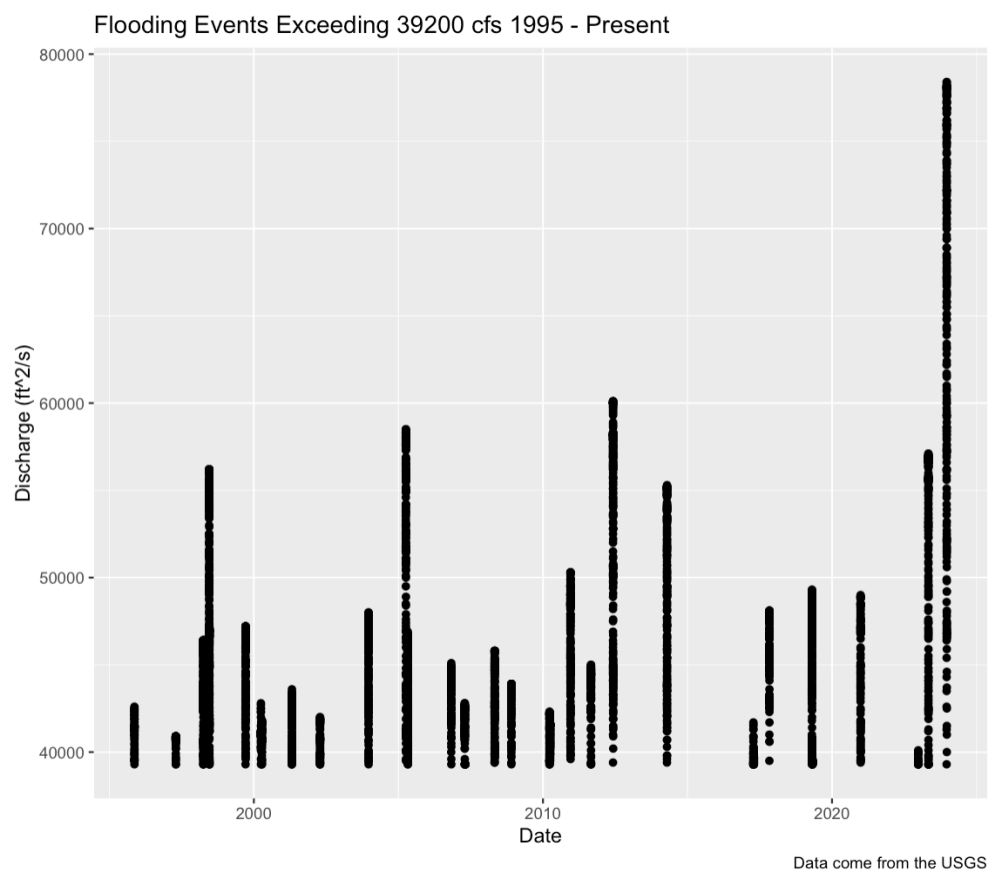
Flooding events can be established in a variety of different ways. That could be through impact assessment or meteorological data. Additionally, hydrologists often establish flooding events by identifying events that surpass a certain discharge threshold. In this investigation, the threshold set was the 99.5 percentile of the twenty-five-year average data calculated earlier (lines 171-172). This was solved to be 39200. This was then plotted to see if these flooding events have increased.

Study Findings



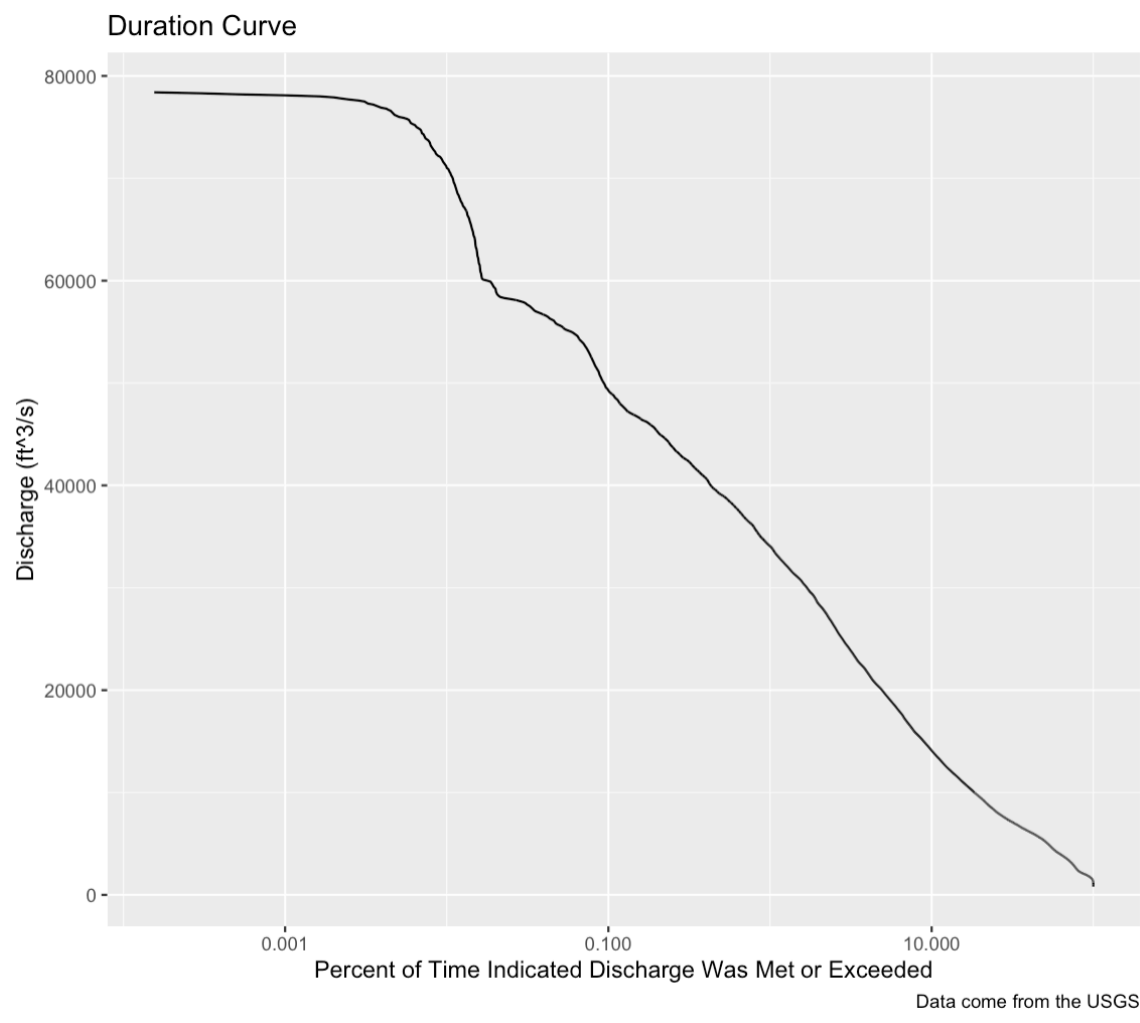
The overlaid plot of the 25-year averaged discharge plotted over the most recent four calendar years provided an interesting insight into the flow patterns. It quickly becomes evident that flow peaks in the spring (April - May), retreats in the summer, and slightly increases in the late Fall (November - December). Each year has distinct characteristics. 2021 was a very dry year, particularly in the summer. The plot clearly demonstrates that 2023 has consistent flooding events (identified by extreme peaks in the graph). Particularly, there is a significant flooding event in May, and the December event discussed in the intro is very apparent. The peak discharge is remarkably higher than all other data points. 2023 also demonstrates the wet summer that all residents in the surrounding area experienced. Data

from 2020, 2022, and 2023 (2021 was dry in its entirety) demonstrate that discharges are higher with more flooding events in the early winter (late December). Each of these years has flooding events extreme for the small peak in discharge demonstrated in the plot of the 25-year average. This could attributed to the shorter winter weather. Temperatures remain elevated for longer; thus, the precipitation in the early winter is primarily rain rather than snow.



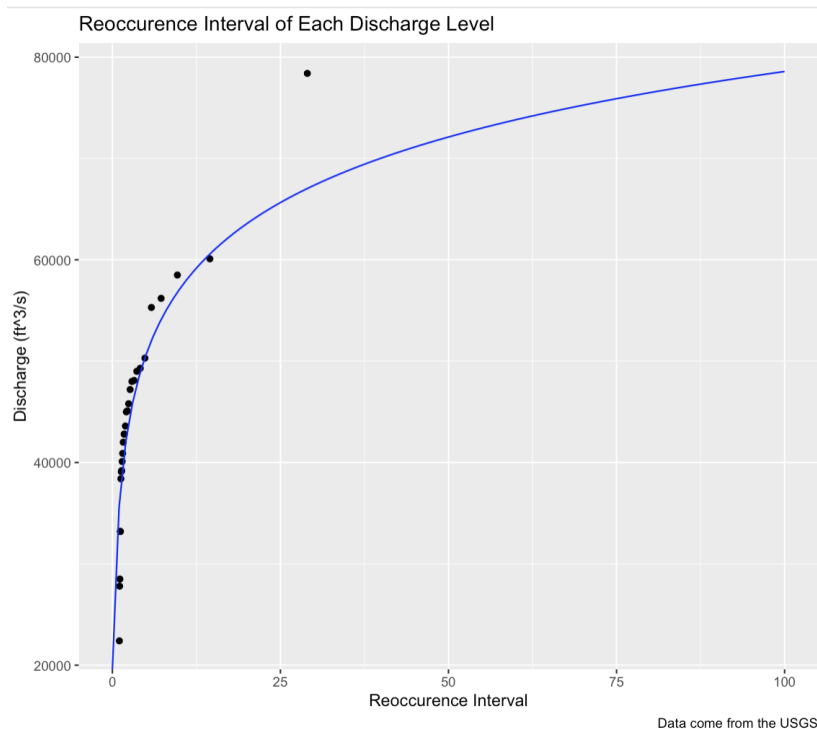
This plot demonstrates the flooding events (discharges greater than 39200 ft²/s). It resembles a bar plot because numerous data points are taken during a flooding event. For example, if a flooding event occurred over 24 hours, that includes 96 data points. So, each bar represents a distinct flooding event. The December 2023 flooding event is more extreme

than the others, given the remarkable discharge level. The plot does not indicate that the frequency of flooding events has increased in the past 29 years. A theoretical data set with more years of data would permit better conclusions to be made.



This duration curve demonstrates the frequency each discharge was met or exceeded. This plot demonstrates that the flooding events of late 2023 were an extreme rarity. The plateau on the graph at 78,000 is directly from that event. This can be assumed because no event in the database is close to that extreme discharge. You can see that events at such discharge occur less than 0.001% of the time. The graph should have a

logarithmic relationship like all other rivers (linear in this graph because the x-axis axis logarithmic). However, the scarcity of data points exceeding 77,000 ft²/s interrupts the logarithmic trend. Additionally, the interruption of the logarithmic trend at 60,000 ft²/s is due to some isolated weather trend during the winter 2023 event. There is one isolated data point above 60,000 otherwise, so this recent event completely dictates the plot at higher discharges. Thus, the inconsistent trend is due to the lack of data points and a localized trend to the winter 2023 storm event. There is a chance that the break in trend could be due to equipment malfunction at that discharge level as well.



This reoccurrence plot effectively demonstrates the logarithmic behavior of discharge rates, aside from the extreme outlier of the December 2023 event. That event is plotted with a reoccurrence interval of 29 years because it occurred once in the 29-year history. But realistically, when you refer to the line of best, it is a 97-year flood. Using the

equation retrieved from the model, the hundred-year flood, 50-year flood, 25-year flood, and 5-year flood were solved. The results are shown in the code below. A one-hundred-year flood is 78,585.3 ft²/s, very similar to the December event. Whereas a 5-year flood is 50,640.36. The recurrence interval for the December 2023 event was solved to be a 96.99-year flood. Given the RI's proximity to 100, hydrologists, meteorologists, and newscasters have referred to that event as a hundred-year flood.

```
123 # Hundred Year Flood
124 hundred_year_flood <- 35627.12 + (9328.25 * log(100))
125 print(hundred_year_flood)
126 # Hundred Year Flood = 78585.3
127
128 # 50 Year Flood
129 fifty_year_flood <- 35627.12 + 9328.25 * log(50)
130 print(fifty_year_flood)
131 # 50 Year Flood = 72119.45
132
133 # 25 Year Flood
134 twenty_five_year_flood <- 35627.12 + 9328.25 * log(25)
135 print(twenty_five_year_flood)
136 # 25 Year Flood = 65653.6
137
138 # 5 Year Flood
139 five_year_flood <- 35627.12 + 9328.25 * log(5)
140 print(five_year_flood)
141 # 5 year Flood = 50640.36
142
143 # December 2023 Event
144 # 78300 = 35627.12 + 9328.25 * log(x)
145 # x = 96.99
146 # 96.99 Year Flood
```

Conclusions

This investigation provided informative general context for the December 2023 flooding event. It was found that 2023 had many more extreme flow events (found in the overlay plot) and higher flows in general. The Duration Curve and the Reoccurrence Interval plot demonstrate how extreme the December 2023 event was. Such discharges (78300 ft³/s) occur less than 0.001% of the time and occur once every 96.99 years. This event was

concerning. However, it appears (Flood Event Plot) that such events are not becoming more frequent. Moving forward, flooding events in the Androscoggin and rivers elsewhere should continue to be monitored. As climate change continues to become more extreme in the coming years, we may see an increase in flooding events from intense meteorological phenomena.